

Power BI Essentials

A Beginner's Tutorial

Using the Apocalypse Food Prep dataset

Topics covered:

Power Query • Joins & Relationships • Drill Down
DAX • Groups

Introduction

Microsoft Power BI is a business analytics platform that lets you connect to data from many sources, transform it into a clean model, and build interactive reports and dashboards from it. This tutorial walks you through five fundamental skills every Power BI user needs: shaping data with Power Query, modeling data with relationships, exploring it with drill down, calculating with DAX, and organizing it with groups.

The exercises in this tutorial use the Apocalypse Food Prep dataset. The dataset tracks the prices of survival supplies (rice, beans, bottled water, flashlights, etc.) across three stores (Walmart, Costco, and Target) over four months in 2022. A second set of files extends the scenario into a small business view with three related tables: products being sold, the sales transactions, and the customers placing the orders.

What you will need

- Power BI Desktop installed on your computer (free download from Microsoft).
- The five Excel files provided with this tutorial.

How this tutorial is organized

Each of the five lessons starts with a short concept explanation, then walks through the exact clicks needed to perform the technique on the sample data. After the five lessons you will find ten hands-on tasks that combine the techniques. Work through them in order — each task assumes the previous lessons have been completed.

Tip

Save your Power BI file (.pbix) often. Power BI Desktop does not auto-save by default, and rebuilding a model from scratch is no fun.

1. Power Query

Power Query is the data preparation engine inside Power BI. It is where you clean, reshape, and combine data before it is loaded into your model. Anything you do in Power Query is recorded as a step that re-runs automatically every time the data is refreshed, so you only have to fix a problem once.

Common things you do in Power Query include changing data types, removing blank rows, splitting one column into several, replacing values, trimming extra spaces, and merging or appending tables. The Power Query Editor opens in a separate window from the main Power BI report view.

Loading the data

1. Open Power BI Desktop and click Home, then Get data, then Excel workbook.
2. Browse to Apocalypse_Food_Prep_-_Power_Query_Editor_Tutorial.xlsx and click Open.
3. In the Navigator pane, tick the **Purchase Tracker** sheet.
4. Click Transform Data (not Load) to open the Power Query Editor.

Why Transform Data and not Load?

Load brings the data straight into the report exactly as it is in the file. Transform Data opens the Power Query Editor first so you can clean the data before it ever reaches your model. For raw source files, always go through Transform Data.

Promoting headers and checking data types

When the data first opens, look at the column headers. If they show as Column1, Column2 and the real header is in row one, click Home then Use First Row as Headers. The Purchase Tracker sheet already has clean headers, so you only need to verify that each column has the right type.

- **Store** — should be Text (icon: ABC).
- **Product** — should be Text.
- **Price** — should be Decimal Number (icon: 1.2).
- **Date** — should be Date. If it shows as a whole number, click the icon to the left of the column name and change the type to Date.

About Excel serial dates

The raw file shows dates like 44562 instead of 1/1/2022. That is the Excel serial date number. Power Query converts it correctly when you set the column type to Date.

Cleaning the Product column

Look closely at the Product column. The value Bottled Water has a trailing space — the trailing space is a common source of bugs because Bottled Water and "Bottled Water " look identical but Power BI treats them as different products. Power Query can fix this in one click.

1. Click the Product column header to select the column.
2. On the Transform tab, click Format, then Trim. This removes leading and trailing whitespace.
3. Click Format, then Clean to strip out any non-printable characters.

Replacing values

Suppose you want to rename Walmart to Wal-Mart everywhere. Right-click the Store column and choose Replace Values. Enter Walmart in the first box and Wal-Mart in the second. Click OK. The change is applied to every row, and a new step is recorded in the Applied Steps panel on the right.

Adding a calculated column in Power Query

Power Query can also add new columns. The Add Column tab has options for Custom Column (M-code formulas), Conditional Column (if/then logic), and Index Column (a row number). For example, to add a column that flags expensive products:

1. Click Add Column, then Conditional Column.
2. Name the new column **Price Tier**.
3. Add the rule: if Price is greater than 20, output High. Otherwise output Low.
4. Click OK. A new column appears at the right of the table.

Applied Steps and the M language

Every action you take is recorded in the Applied Steps pane on the right side of the Power Query Editor. You can click any step to see the data as it was at that point, delete a step with the X next to it, or reorder steps by dragging. Behind the scenes, each step is a line of M code, the formal language Power Query uses. You normally never need to write M by hand, but the formula bar above the data lets you read and edit it if you want to.

Loading the cleaned data

When you are happy with the cleaned table, click Home then Close & Apply. Power BI loads the transformed data into the model and returns you to the report view.

Key takeaway

Power Query is repeatable. Anything you fix here is applied automatically every time you refresh the data, so future versions of the source file get the same treatment without any extra work.

2. Joins and Relationships

Real datasets are almost never a single big table. Instead, the same information is spread across smaller tables that connect to one another through shared key columns. Power BI gives you two ways to combine tables: a join in Power Query, which physically combines rows into one wide table, and a relationship in the model, which keeps the tables separate but lets visuals pull from all of them at once.

Joins (Power Query)

A join in Power Query is a one-time merge that produces a new combined table. Use a join when you want a flattened result for export, or when the relationship is so simple that you do not want a separate table at all.

1. In Power Query, click Home, then Merge Queries.
2. Pick the second table to merge with the current one.
3. Click the matching key column in each table.
4. Choose a join kind: Left Outer keeps every row from the first table, Inner keeps only matches, and so on.
5. Click OK, then expand the new column to bring in the fields you need.

Six join kinds at a glance

Left Outer — every row from the first table plus matches from the second.

Right Outer — every row from the second table plus matches from the first.

Full Outer — every row from both tables.

Inner — only rows that have a match in both tables.

Left Anti — rows in the first table that do not match.

Right Anti — rows in the second table that do not match.

Relationships (the data model)

Relationships are the standard way Power BI links tables. Tables stay separate and can be used independently, but a visual that mixes fields from two related tables works correctly because Power BI follows the relationship behind the scenes. This is the approach used in almost every real Power BI report.

Walking through the Relationship Tutorial file

Open `Apocalypse_Food_Prep_-_Relationship_Tutorial.xlsx`. This file contains three tables: **Apocalypse Store** (a product catalog), **Apocalypse Sales** (one row per order), and **Customer Information** (one row per customer).

1. In Power BI Desktop, click **Get data**, then **Excel**, and select all three sheets in the Navigator.
2. Click Load. All three tables appear in the Data pane on the right.
3. Click the Model view icon on the left side of the screen (it looks like three connected boxes).
4. Power BI may have auto-detected relationships. If it has, you will see lines connecting the tables. If not, you create them manually.

Creating a relationship manually

1. In Model view, click and drag `Product ID` from the `Apocalypse Sales` table onto `Product ID` in the `Apocalypse Store` table.
2. Drag `Cust ID` from `Apocalypse Sales` onto `Customer ID` in `Customer Information`.
3. A line appears between each pair of tables. Hover over a line to see which columns are joined.

Cardinality

Cardinality describes how rows on one side of a relationship match rows on the other. The two you will use 95 percent of the time are:

- **One-to-many (1:*)** — the most common type. One row in the lookup table matches many rows in the fact table. In this dataset, one product in `Apocalypse Store` can appear on many sales rows in `Apocalypse Sales`. The arrow on the relationship line points from the one side to the many side.
- **Many-to-many (*:*)** — use only when neither table has unique keys. Many-to-many is powerful but can produce confusing totals if you are not careful, so avoid it unless you really need it.

Cross-filter direction

Each relationship has a filter direction, shown by the arrow on the line. Single direction (the default) means filters flow only from the one side to the many side: filtering `Apocalypse Store` filters `Apocalypse Sales`, but not the other way around. Both directions lets filters flow in either direction. Stick with single direction unless you have a clear reason to change it — it keeps your model predictable.

Editing a relationship

Double-click a relationship line, or go to Home then Manage Relationships, to edit it. From there you can change the joined columns, the cardinality, the filter direction, or deactivate the relationship without deleting it.

Joins vs Relationships — when to use which

Use a join in Power Query when you only need a single combined table, you are exporting flat data, or the join is part of cleaning a single source.

Use a relationship when you have proper fact and lookup tables, you want to use them in many different visuals, and you want filters to flow between them automatically.

3. Drill Down

Drill down lets a viewer of your report click on a visual to see more detail, then click again to see even more. The classic example is a date hierarchy: a chart starts by showing yearly totals, you drill down once to see quarters, again to see months, and again to see individual days. Drill down keeps reports compact while still allowing deep exploration.

How hierarchies work

A hierarchy is a stack of fields that represent levels of detail. Power BI automatically builds a date hierarchy (Year, Quarter, Month, Day) for any column with the Date type. You can also build your own hierarchies — for example, in this dataset you could create a Location hierarchy of Store, then Product.

Building a drill-down chart

1. Open `Apocalypse_Food_Prep_-_Drill_Down_Tutorial.xlsx` in Power BI.
2. In the Report view, drag a **Clustered column chart** onto the canvas.
3. Drag the Date field onto the X-axis. Power BI replaces it with the Date hierarchy automatically (Year, Quarter, Month, Day).
4. Drag Price onto the Y-axis. By default it sums the prices.
5. The chart now shows totals by year. Look at the four small icons that appear at the top corner of the visual.

The four drill icons

- **Drill Up (upward arrow)** — moves you back up one level in the hierarchy.
- **Drill Down on a single value (single downward arrow)** — click this to turn drill mode on, then click a single bar to drill into just that value.
- **Go to the next level (split arrows)** — moves the entire chart one level down. The chart now shows the next level for all values, not just one.
- **Expand all down one level (double down arrows)** — shows both the current and the next level together (for example, Year and Quarter combined on the X-axis).

Building your own hierarchy

You can group related fields into a custom hierarchy so they always travel together.

1. In the Data pane, right-click the Store field and choose Create hierarchy.
2. Drag the Product field onto the new Store Hierarchy.
3. Use the hierarchy in a chart. You can now drill from Store down to Product just like you did with dates.

Drill through (a related concept)

Drill through is different from drill down. With drill through, right-clicking a value sends the user to a separate report page that has been built specifically to show details for that value. Drill through is great for jumping from a summary dashboard to a detailed page filtered to one customer or one product. You set it up by adding a Drill through field on the destination page.

Tip

Add a Back button on any drill-through page so users can easily return. Insert ribbon, then Buttons, then Back.

4. DAX

DAX, short for Data Analysis Expressions, is the formula language used in Power BI to create custom calculations. It looks similar to Excel formulas but works on whole tables and columns instead of single cells. DAX is what you use whenever the field you need does not already exist in the data.

Calculated columns vs measures

There are two main types of DAX formulas. Knowing when to use each one is the single most important DAX skill for beginners.

- **Calculated column** — a new column added to a table. The formula runs once for every row when the data is loaded, and the result is stored in the model. Use a calculated column when the value belongs on the row itself, such as a Profit column on a Sales table.
- **Measure** — a calculation that is evaluated dynamically based on whatever filters and slicers are applied to the visual using it. Measures do not store values; they recalculate on the fly. Use a measure for sums, averages, ratios, and any total that should respond to user filtering.

Rule of thumb

If you are not sure which to use, start with a measure. Measures are more flexible, use less memory, and respond to slicers automatically. Reach for a calculated column only when you need a per-row value that other formulas or visuals will use.

Creating a measure

1. In the Data pane, right-click a table and choose New measure.
2. Type the formula in the formula bar at the top.
3. Press Enter. The new measure appears in the table with a calculator icon.

Useful DAX functions to learn first

- **SUM(column)** — adds up all values in a column.
- **AVERAGE(column)** — average of a column.
- **COUNT(column)** and **COUNTROWS(table)** — count values or rows.
- **MIN** and **MAX** — smallest and largest values.
- **DIVIDE(numerator, denominator, alternate)** — safe division that returns the alternate value (default 0) when the denominator is zero.
- **CALCULATE(expression, filter1, filter2, ...)** — the most powerful DAX function. Re-evaluates an expression with the given filters applied.
- **RELATED(column)** — looks up a value from a related table on the one side of a relationship.

- `SUMX(table, expression)` — evaluates the expression once per row and adds up the results. Useful when the formula must run per row before being summed.

Examples using the Apocalypse data

Open the Relationship Tutorial file and add the following measures to the Apocalypse Sales table.

Total Units Sold — sums the Units Sold column.

```
Total Units Sold = SUM('Apocalypse Sales'[Units Sold])
```

Total Revenue — multiplies units by the related product price, then sums. Note the use of `RELATED` to pull the price from the products table.

```
Total Revenue = SUMX('Apocalypse Sales', 'Apocalypse Sales'[Units Sold] *  
RELATED('Apocalypse Store'[Price]))
```

Total Cost — same idea, using Production Cost.

```
Total Cost = SUMX('Apocalypse Sales', 'Apocalypse Sales'[Units Sold] *  
RELATED('Apocalypse Store'[Production Cost]))
```

Profit — revenue minus cost. Notice how a measure can reference other measures.

```
Profit = [Total Revenue] - [Total Cost]
```

Profit Margin — profit as a percentage of revenue, with safe division.

```
Profit Margin = DIVIDE([Profit], [Total Revenue], 0)
```

After creating Profit Margin, click it in the Data pane and use the Format options to display it as a percentage.

Calculated column example

On the Apocalypse Store table, add a calculated column called **Markup** that computes the markup amount per product:

```
Markup = 'Apocalypse Store'[Price] - 'Apocalypse Store'[Production Cost]
```

This is a calculated column rather than a measure because the value belongs on the product row itself and does not depend on filters. To create it, right-click the Apocalypse Store table and choose New column.

Common beginner mistake

Putting the same formula into a calculated column when it should have been a measure. If the result is something you want to total, average, or filter dynamically (like Total Revenue), it should be a measure. Calculated columns are for per-row attributes only.

5. Groups

Groups let you bundle individual values together into bigger categories. They are useful when the raw data is too granular for a clean visual — for example, ten products are too many for a clean pie chart, but three categories like Food, Tools, and Equipment fit perfectly. Power BI offers two flavors of grouping: list groups for discrete values, and bins for numeric or date ranges.

List groups (categorical)

List groups bundle text values together. With the Bins/Lists Tutorial file open, follow these steps to group the ten survival products into a smaller set of categories.

1. Open `Apocalypse_Food_Prep_-_Bins_Lists_Tutorial.xlsx` and load both sheets.
2. In the Data pane, right-click the `Product Name` field and choose **New group**.
3. In the Groups dialog, set **Group type** to **List**.
4. In the Ungrouped values box, hold **Ctrl** and click `Nylon Rope`, `Duct Tape`, and `Multitool Survival Knife`. Click **Group**.
5. Double-click the new group label and rename it to **Tools**.
6. Repeat to create a **Safety Gear** group with `N95 Mask` and `Weatherproof Jacket`.
7. Repeat to create a **Survival Equipment** group with the remaining items.
8. Tick **Include Other** group to put any future products into a default bucket. Click **OK**.

A new field called `Product Name (groups)` appears in the Data pane. Use it instead of `Product Name` in your visuals to roll the products up into your three categories.

Bins (numeric or date)

Bins divide a continuous range into equal-size buckets. Use them when you want a histogram or a clean breakdown of a numeric column.

1. Right-click the `Price` field on `Apocalypse Store` and choose **New group**.
2. Set **Group type** to **Bin**.
3. Choose **Bin type**. **Size of bins** lets you set a fixed bin width (for example, 10 to get the buckets 0–10, 10–20, 20–30, and so on). **Number of bins** lets you specify how many buckets you want and Power BI works out the size.
4. Set the bin size to 10 and click **OK**.

A new field called `Price (bins)` is created. Use it on the X-axis of a column chart with a count of products on the Y-axis to make a quick price-distribution histogram.

Editing or deleting a group

Right-click the group field and choose Edit groups to add or remove members and rename categories. Choose Delete to remove the group entirely. Editing a group automatically updates every visual that uses it.

When to use groups vs DAX

Groups are quick, click-driven, and ideal for small static categorizations like the ten-product example above. If your grouping rule is complex (for example, group by price band AND store), use a calculated column with a SWITCH or IF statement in DAX instead.

Practice Tasks

The ten tasks below build directly on the lessons. Work through them in order — each one assumes you have completed the earlier tasks. The goal is to give you confidence using all five techniques together on the Apocalypse Food Prep dataset.

Task 1 — Power Query: Clean the purchase tracker

Open `Apocalypse_Food_Prep.xlsx` in Power BI through Transform Data. In the Power Query Editor, do all of the following:

- Trim the Product column to remove the trailing space on Bottled Water.
- Set the Date column to the Date data type.
- Rename the query to Purchases.
- Click Close & Apply to load the cleaned data.

Task 2 — Power Query: Add a Price Tier column

Reopen the Power Query Editor (Home, then Transform data) and add a Conditional Column called Price Tier with these rules:

- If Price is greater than or equal to 20, output High.
- Else if Price is greater than or equal to 5, output Medium.
- Otherwise, output Low.

Apply the changes and confirm the new column appears in the Data pane.

Task 3 — Relationships: Build the three-table model

Open a new Power BI file and load all three sheets from `Apocalypse_Food_Prep_-_Relationship_Tutorial.xlsx`. In Model view:

- Create a one-to-many relationship from Apocalypse Store to Apocalypse Sales on Product ID.
- Create a one-to-many relationship from Customer Information to Apocalypse Sales on Customer ID / Cust ID.
- Confirm both relationships have single cross-filter direction (arrow pointing toward Apocalypse Sales).

Task 4 — DAX: Three core measures

On the Apocalypse Sales table from Task 3, create the following three measures:

```
Total Units Sold = SUM('Apocalypse Sales'[Units Sold])
Total Revenue = SUMX('Apocalypse Sales', 'Apocalypse Sales'[Units Sold] *
RELATED('Apocalypse Store'[Price]))
Total Cost = SUMX('Apocalypse Sales', 'Apocalypse Sales'[Units Sold] *
RELATED('Apocalypse Store'[Production Cost]))
```

Verify each measure shows a sensible total when dropped into a card visual.

Task 5 — DAX: Profit and profit margin

Add two more measures to the Apocalypse Sales table that reference the measures you built in Task 4:

```
Profit = [Total Revenue] - [Total Cost]
Profit Margin = DIVIDE([Profit], [Total Revenue], 0)
```

Format Profit and Profit Margin appropriately (currency for Profit, percentage for Profit Margin) using the Format options in the Measure tools ribbon.

Task 6 — Drill Down: A drillable revenue chart

On a new report page, add a clustered column chart and configure it as follows:

- X-axis: Date Purchased (Power BI will use the auto-generated date hierarchy).
- Y-axis: the Total Revenue measure from Task 4.

Use the four drill icons in the top corner of the chart to drill from Year down to Quarter, then Month, then Day. Take a screenshot at each level so you can see the difference.

Task 7 — Groups: Categorize the products

On the Apocalypse Store table, create a list group on Product Name that bundles the ten products into three categories:

- **Tools** — Nylon Rope, Duct Tape, Multitool Survival Knife, Stainless Steel Axe.
- **Safety Gear** — N95 Mask, Weatherproof Jacket.
- **Survival Equipment** — Backpack, Solar Battery Flashlight, Water Purifier, Waterproof Matches.

Build a pie or donut chart that uses your new Product Name (groups) field as the Legend and Total Revenue as the Values to see which category drives the most revenue.

Task 8 — Groups: Price bins on the products

Create a Bin group on the Price column of Apocalypse Store. Use a bin size of 10 and name the field Price Range. Build a column chart with this new field on the X-axis and a count of products on the Y-axis. The result is a quick histogram of the product price distribution.

Task 9 — Drill down on a custom hierarchy

Create a custom hierarchy on the Customer Information table:

- Right-click State and choose Create hierarchy.
- Drag City and then Customer into the new hierarchy.

Build a bar chart using this hierarchy on the Y-axis and Total Revenue on the X-axis. Drill down through State, City, and Customer to see how revenue rolls up geographically.

Task 10 — Build a one-page dashboard

Bring everything together on a single report page. The page must include all of the following:

- Three card visuals at the top showing Total Units Sold, Total Revenue, and Profit Margin.
- A drillable column chart of Total Revenue over time.
- A pie chart of Total Revenue by Product Name (groups) — the categories from Task 7.
- A bar chart of Total Revenue by State, using the hierarchy from Task 9.
- At least one slicer (for example, on Customer or on Date) so a viewer can filter the page interactively.

Save the file as Apocalypse_Dashboard.pbix. Click any value on any visual to confirm the others react correctly. If they do, your relationships, measures, groups, and drill-downs are all wired together correctly.